Isotope Effects

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This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle* Plots were typeset using *MatPlotLib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

Carbon-12 and carbon-14 ($^{12}_{6}C$ & $^{14}_{6}C$)are the same element. Hydrogen and deuterium ($^{1}_{1}H$ & $^{2}_{1}H$) are also the same element. Each pair has the exact same nuclear charge and electronic arrangement. They are identical in every way – except in mass. Let us explore how mass affects rates and equilibria in chemistry and how these observations can be used as tools to investigate mechanisms.[1]

[1] Read chapter 8.1.

Some Definitions to Complete Before Class

1. What energy regime do isotope effects operate within?
2. Describe primary isotope effects and secondary isotope effects and how they differ.
3. Describe the difference between equilibrium and kinetic isotope effects.
4. Describe an inverse isotope effect and how it can occur.

Some Math to Make the Class Meeting More Fun

Some equations that may be helpful.

$$\bar{\nu} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$E = hc\bar{\nu}$$

$$k = Ae^{\frac{-E_a}{RT}}$$

5. Using the data in figure 1, calculate the force constant of a C – H sp^3 bond.
6. Now use this force constant to calculate the expected frequency of vibration of the equivalent C – D bond.
7. Compare your result to the experimental data in figure 2.
8. Calculate the energy difference for complete bond breaking between the case of a C – H and a C – D sp^3 bond using the data from the above calculations.
9. Calculate the difference in rate for bond breaking, k_H/k_D (assuming the transition state is nearly identical in energy to the products. Is this the maximum value for a primary kinetic isotope effect?)
10. Consider what you have learned in IR spectroscopy and explain why the large peak near 1060 cm^{-1} is unchanged between the normal and deuterated spectra.
11. Do you think that the pH value measured using a glass electrode in an NMR tube is accurate?[2]

[2] "Use of Glass Electrodes to Measure Acidities in Deuterium Oxide." P.K. Glasoe, F.A. Long, *J. Phys. Chem.*, **1960**, *64*, 188–190. https://doi.org/10.1021/j100830a521

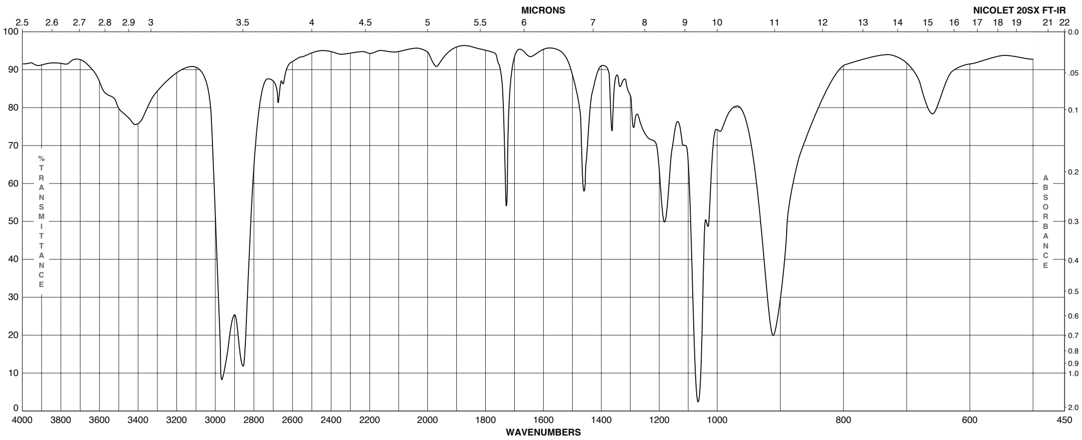


Figure 1: FTIR spectrum of tetrahydrofuran. From the Spectral Database for Organic Compounds at https://sdbs.db.aist.go.jp/sdbs/cgi-bin/landingpage?sdbsno=497

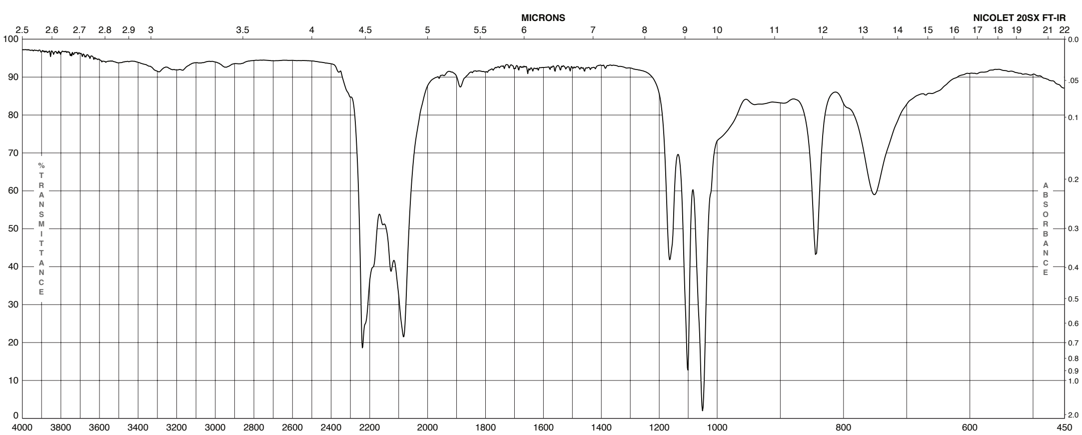


Figure 2: FTIR spectrum of deuterated tetrahydrofuran-d_8. From the Sigma-Aldrich catalogue at https://www.sigmaaldrich.com/CA/en/product/aldrich/184314

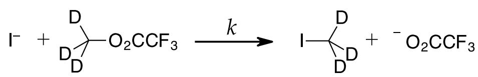
Class Exercises Using Isotope Effects

We won't get to all of these. We might not get to any of them in the class time we have. Be sure to try them on your own and ask questions.

12. Predict the type and magnitude of the kinetic isotope effects for the reactions listed in table 1 with substitutions of the indicated protons with deuterium? Write a brief defence of each decision.

Table 1: Cases for problem #12.

a)



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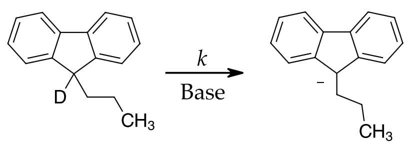

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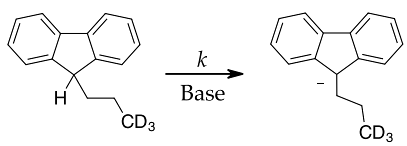

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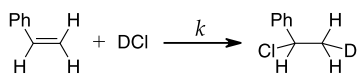
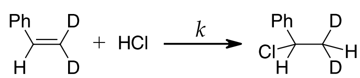


e)



13. For the following reaction, propose structures for the transition state in the rate-determining step that would be consistent with the isotope effects listed in table 2. Indicate which bonds are breaking or forming in the T.S. of each reaction. What type of reaction describes each mechanism (i.e. S_N1, etc.). Which of the three set of data below is likely observed in this reaction and which two sets are likely fictional?

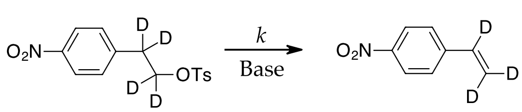


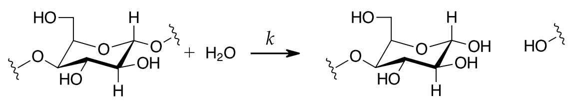
Table 2: Cases for problem #13.

a) $k_H/k_D = 5.2$
$k_{O_{16}}/k_{O_{18}} = 1.013$

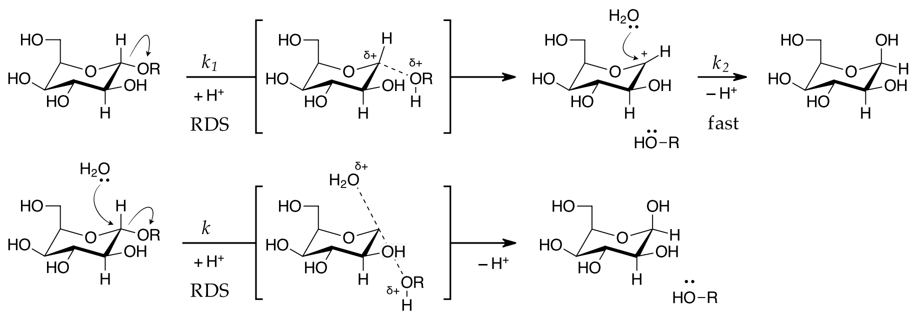
b) $k_H/k_D = 1.3$
$k_{O_{16}}/k_{O_{18}} = 1.016$

c) $k_H/k_D = 4.3$
$k_{O_{16}}/k_{O_{18}} = 1.000$

14. For the glycosidic cleavage of the bacterial cell wall catalyzed by *lysozyme*, the following reaction occurs...



It is proposed that the reaction proceeds by one of two mechanisms...



(a) Describe each mechanism (or each step in each mechanism) using the common designations (i.e. S_N1, etc.)

(b) Design an experiment that involves isotopic substitution of hydrogen atoms that will elucidate the mechanism. Describe the isotopic substitutions you would make.

(c) Describe the expected results for your experiment for each mechanism.

(d) In addition to your experiments above, explain what the expected primary kinetic isotope effect would be for isotopic substitution of the oxygen in water in each scheme.

(e) Explain what the expected primary kinetic isotope effect would be for isotopic substitution of the oxygen in the hydroxyl leaving group in each scheme.

Textbook Questions

As you read about isotope effects, try these textbook problems: Ch. 8: 2, 3, 13, 19 & 20.

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will…

- be able to express the definitions involved in isotope effects
- understand the theory of isotope effects and be able to apply it in calculations.
- be able to predict the magnitude of an isotope effect in an experimental design and interpret experimental results.
- have reviewed vibrational spectroscopy (a bit)
- now understand that electronic energy is not the only energy that affects reaction rates and equilibrium.
- be able to accurately measure the protonating power (pD) in D_2O using a pH electrode.
- **Computer Skill**: be able to use an interactive *Python* notebook to perform and document calculations.